MIL-V-46727C(AT)
10 October 1986
SUPERSEDING
MIL-V-46727B(AT)
23 May 1980

MILITARY SPECIFICATION

VALVE, THREE-POSITION, FOUR-WAY HYDRAULIC, MANUAL

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of manually operated, three-position, four-way, hydraulic directional control valves with an operating pressure of 3000 pounds per square inch (psi) (see 6.1).

1.2 Classification. Valves shall be of the following types, as specified (see 6.2)

Type I	- Rated flow capacity: 6 gallons per minute (gpm).
Type II	- Rated flow capacity: 2 gpm.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2590

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
MILITARY

MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-V-5529	- Valves, Hydraulic Directional Control, Rotary Selector.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base for Preservation and Operation.

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services Section 2 Unified Inch Screw Threads-UN and UNR Thread Forms.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government-documents, drawings, and publications. The following other Government documents, drawings, and publications form apart of this specification to the extent specified herein. Unless otherwise, specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

9094352	- Valve, Three Position, Four Way (Type I).
9098279	- Valve, Three Position, Four Way (Type II).

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Method of Salt Spray (fog) Testing.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish valves which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the valves to be furnished to the Government. All subsequent valves delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall conform to the applicable requirements of MIL-V-5529. Material not specifically designated shall be of such quality as to assure conformance of the valve to requirements specified herein and in referenced specifications, standards, and drawings. Material shall be free from defects which adversely affect performance of serviceability of the finished product. Exposed functional parts shall be fabricated from suitable corrosion resistant materials or treated to prevent corrosion (see 4.7.1 and 6.3).

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3.3 Design and construction. Unless otherwise specified (see 6.2), the valve shall be designed and constructed to the form and dimensions specified herein, and on the applicable drawing specified below (see 4.7.1 and 4.7.2):

Type I	- Drawing 9094352.
Type II	- Drawing 9098279.

3.3.1 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.7.1).

3.3.2 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Subassemblies, composed of mating components, shall be interchangeable as assembled units (see 4.7.1).

3.3.3 Threaded parts. Screw threads shall conform to FED-STD-H28/2. The form, number per inch, and class of threads shall be as specified on the applicable drawings (see 4.7.1 and 4.7.2).

3.3.4 Valve spool. Valve spool stems exposed to dirt and moisture shall be provided with wipers and seals meeting the life requirements specified herein. All exposed control-valve-spool surfaces shall be hardened to Rockwell C 50 to 65. The spool surface shall be plated to resist corrosion. After plating, maximum spool surface roughness shall not exceed 16 root mean square (rms). The plating shall be such that the spool shall not corrode when tested for a not less than 30 hours, as specified in ASTM B117 (see 4.7.3.11.1). The spool shall be hydraulically balanced so that forces required to manipulate the levers shall not vary more than 10 percent (%) due to variations in pressure and flow. Spools shall not bind under any operating conditions specified herein (see 4.7.1 and 4.7.2).

3.3.5 Weight. The gross weight of the valve shall not exceed 2.0 pounds (lb) (see 4.7.2).

3.4 Performance.

3.4.1 Rated flow and pressure drop.

3.4.1.1 Type I. The rated flow capacity of the valve shall be 6 gpm through any interconnected ports, with not more than a 250 psi pressure drop across the valve (see 4.7.3.1).

3.4.1.2 Type II. The rated flow capacity of the valve shall be 2 gpm through any interconnected ports, with not more than a 100 psi pressure drop across the valve (see 4.7.3.1).

3.4.2 Operating torque.

3.4.2.1 Type I. The torque required to operate the valve shall be not more than 52 pound-inches (see 4.7.3.2).

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3.4.2.2 Type II. The torque required to operate the valves shall be not more than 25 pound-inches (see 4.7.3.2).

3.4.3 Operating pressure. The valve shall operate satisfactorily throughout a hydraulic pressure range of 0 to 3000 psi applied to any port, except the return, while using a hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083 (see 4.7.3.3).

3.4.4 Proof pressure. The valve shall withstand a hydraulic proof pressure of 4500 psi, except for the return port which shall withstand 1500 psi, without external leakage, permanent deformation, or failure (see 4.7.3.4).

3.4.5 Burst pressure. The valve shall withstand a hydraulic pressure of 7500 psi, except for the return port which shall withstand 2500 psi, without bursting (see 4.7.3.5).

3.4.6 Internal leakage. The valve shall withstand application of 3000 psi hydraulic pressure to sealing members of the individual fluid chambers without external leakage, and with a combined internal leakage from the ports not exceeding 10 cubic centimeters per minute (cc/min) (see 4.7.3.6).

3.4.7 Endurance. The valve shall withstand 20000 cycles of operation at 3000 psi hydraulic working pressure, without external leakage. A cycle shall consist of an operation of the valve from one extreme position to the other extreme position and return to the starting position. The valves shall subsequently meet the requirements of 3.4.6 (see 4.7.3.7).

3.4.8 Environmental.

3.4.8.1 High temperature. The valve shall withstand exposure to a high temperature of 160 $\pm$ 5°F in accordance with method 501, procedure II of MIL-STD-810, without damage and shall conform to the performance requirements of 3.4.1 and 3.4.7. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.8).

3.4.8.2 Low temperature. The valve shall withstand exposure to a low temperature of -65 (+5, -0) ^{0}F in accordance with method 502, procedure II of MIL-STD-810, without damage and shall conform to the performance requirements of 3.4.1 and 3.4.7. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.9).

3.4.8.3 Fungus. The valve shall withstand exposure to fungus in accordance with method 508 of MIL-STD-810. After having been subjected to conditions favorable to fungus growth as specified, the valves shall evidence no microbial growth that would adversely affect performance. Subsequent performance shall conform to 3.4.1,3.4.6, and 3.4.7 (see 4.7.3.10).

3.4.8.4 Salt spray. The valve shall withstand exposure to salt spray in accordance with MIL-STD-810, method 509, the duration of test shall be 100 hours. The valve shall evidence no corrosion after being subjected to the salt spray tests. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.11).

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3.4.8.4.1 Valve spool corrosion. The valve spool shall exhibit no corrosion after exposure to salt spray as specified in ASTM B117 for not less than 30 hours (see 4.7.3.11.1).

3.4.8.5 Sand and dust. The valve shall withstand exposure to sand and dust in accordance with method 510, procedure I of MIL-STD-810, except the temperature shall be not more than 125°F for 24 hours without damage. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.12).

3.4.8.6 Shock. The valve shall withstand exposure to shock in accordance with method 516, procedure I of MIL-STD-810, without damage or loosening of components. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.13).

3.4.8.7 Vibration. The valve shall withstand vibration in accordance with method 210A of MIL-STD-202, except vibration shall be for 1 hour at 77° ± 15°F and 30 minutes at minus 65° ± 5°F. Vibration shall be applied in the direction of each of the three major axes without damage or loosening of components. Subsequent performance shall conform to 3.4.1, 3.4.6, and 3.4.7 (see 4.7.3.14).

3.5 Marking. The valve shall be marked in accordance with MIL-STD-130 and as specified on the applicable drawing. When specified on the drawing, the nameplate shall be in accordance with MIL-P-514. Plates are to be made of photographic aluminum (sensitized), not less than 0.030 inch thick, with a black background. Marking shall include the following information in the order shown (see 4.7.2):

a. Valve, Manual, Three-position, Four-way
b. Military part number
c. Manufacturer's serial number
d. Manufacturer's name or trademark
e. Contract or order number
f. Date of manufacture
g. US

3.6 Finish. Unless otherwise specified (see 6.2), finish shall conform to the requirements specified on the applicable drawing (see 4.7.2).

3.7 Workmanship. Workmanship shall be of a quality which will assure a product free of burrs, rust, scratches, chips, sharp edges, or other defects which will deleteriously affect serviceability (4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
   1. Preproduction inspection (see 4.4.1).
   2. Initial production inspection (see 4.4.2).

b. Quality conformance inspections (see 4.5)
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 77° $\pm$ 15°F.
b. Barometer pressure 28.5 + 2.0 inches mercury (Hg).
c. Relative humidity 50 $\pm$ 30 percent.

4.4 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified herein. Approval of the first article sample by the Government shall not relieve the contractor of the obligation to supply valves that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

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4.4.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of four valves. Preproduction inspection shall consist of inspection as specified in table I.

4.4.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select four valves, from the first 10 valves produced under the production contract, for initial production inspection. Initial production units shall be inspected as specified in table I.

4.4.3 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result, of the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Exami-nation	Tests
Materials and construction	3.2 thru 3.3.4	4.7.1	X		
Defects (see 4.5.2 and table II)	3.3, 3.3.3, thru 3.3.5, 3.5 thru 3.7	4.7.2		X	
		4.7.3.1	X		X
Rated flow and pressure drop	3.4.1		X		X
Operating torque	3.4.2	4.7.3.2	X		X
Operating pressure	3.4.3	4.7.3.3	X		
Proof pressure	3.4.4	4.7.3.4	X		
Burst pressure	3.4.5	4.7.3.5	X		X
Internal leakage	3.4.6	4.7.3.6			
Endurance	3.4.7	4.7.3.7	X		
High temperature	3.4.8.1	4.7.3.8	X		
Low temperature	3.4.8.2	4.7.3.9	X		
Fungus	3.4.8.3	4.7.3.10	X		
Salt spray	3.4.8.4	4.7.3.11	X		
Valve spool corrosion	3.4.8.4.1	4.7.3.11.1	X		
Sand and dust	3.4.8.5	4.7.3.12	X		
Shock	3.4.8.6	4.7.3.13	X		
Vibration	3.4.8.7	4.7.3.14			

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4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the valves of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 20 consecutively produced assemblies that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.1.3 Sampling for test. Samples for tests shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability not within tolerance (see 3.3).	SIE 1/ or functional
102	Threads not as specified (see 3.3.3).	SIE
103	Valve spool stem wipers or seal missing or defective (see 3.3.4).	Visual
104	Valve spool hardness not as specified (see 3.3.4).	SIE
105	Valve spool surface finish in excess of that specified (3.3.4).	SIE
106	Hydraulic balance of spool not within limits (see 3.3.4).	Functional
107	Binding (see 3.3.4).	Functional

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TABLE. Classification of defects – Continued.

Category	Defect	Method of examination
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	SIE or Functional
202	Weight greater than specified (see 3.3.5).	SIE
203	Marking illegible or incorrect (see 3.5).	Visual
204	Finish not as specified (see 3.6).	Visual
205	Faulty workmanship (see 3.7)	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests. Each valve selected in accordance with 4.5.1.3 shall be subjected to the quality conformance tests specified in table I, in the order specified using an AQL of 1.0 on the basis of percent defective.

4.6 Failure. Failure of any valve to pass any of the specified quality conformance tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 through 3.3.4 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.3.3 through 3.3.5 and 3.5 through 3.7, shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.7.3 Performance. Unless otherwise specified (see 6.2), performance testing shall be conducted under the following conditions:

a. Fluids. All tests shall be performed with hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083. The hydraulic fluid temperature shall be 90° ± 20°F.
b. Cleaning. All grease or foreign material shall be removed from interior parts of the valve before testing.
c. Air removal. Air shall be completely removed from all hydraulic portions of the valves prior to tests.

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4.7.3.1 Rated flow and pressure drop. To determine conformance to 3.4.1, the rated flow of the valve shall be accurately maintained at the applicable value specified in 3.4.1. A manometer, or other accurate means of measuring pressure drop from one port to another, shall be employed. The pressure drop for valves shall be measured and recorded.

4.7.3.2 Operating torque. To determine conformance to 3.4.2, the valve shall be subjected to a hydraulic pressure of 3000 psi at the pressure port, the torque required to operate the valve shall be measured and recorded.

4.7.3.3 Operating pressure. To determine conformance to 3.4.3, the valve pressure shall be applied to each port, except return, and varied throughout a range of 0 to 3000 psi. Operation shall be observed and results recorded.

4.7.3.4 Proof pressure. To determine conformance to 3.4.4, the valve shall be subjected to proof pressures in accordance with 3.4.4 applied at the pressure and return ports for a period of 2 minutes. The valve shall be manually actuated after application of pressures. There shall be no evidence of external leakage, permanent deformation, or failure. The results shall be recorded.

4.7.3.5 Burst pressure. After completion of all other tests. the valve shall be subjected to the pressure specified in 3.4.5 at the pressure ports and return port for a period of 3 minutes at each application. This test shall be performed for all three positions. When testing pressure ports, the return port shall be open. The rate of application of pressure shall not exceed 26 000 psi per minute. The valve need not be operable after these tests. Results shall be recorded to determine conformance to 3.4.5.

4.7.3.6 Internal leakage. To determine conformance to 3.4.6, the valve shall be tested for leakage by applying hydraulic pressure of 3000 psi to each sealing member, of individual fluid chambers, in succession for a 2 minute period each. The pressure shall then be lowered by 5 to 10 psi (gauge) and held for 2 minutes. There shall be no external leakage and the combined internal leakage from each port shall be recorded.

4.7.3.7 Endurance. To determine conformance to 3.4.7, the valve shall be subjected to hydraulic pressure of 3000 psi and a rated flow as specified (see 3.4.1), for 20 000 cycles. Cycling shall be as specified in 3.4.7. The valve shall be observed for evidence of external leakage during cycling. After cycling, internal leakage shall not exceed the value specified in 3.4.6. The valve shall evidence no excessive wear or breakdown and the results shall be recorded.

4.7.3.8 High temperature. To determine conformance to 3.4.8.1, the valve shall be placed within a test chamber and exposed to high temperature as specified in MIL-STD-810, method 501, procedure II, for a time period of 24 hours. After 24 hours of exposure, valve shall

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be charged to 3000 psi hydraulic pressure and specified flow (see 3.4.1) shall be maintained for a period of 4 hours. At the conclusion of the exposure period, the valve shall be tested as specified in 4.7.3.7 (50 cycles only). The valve shall then be cooled to room temperature and tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.7.3.9 Low temperature. To determine conformance to 3.4.8.2, the valve shall be placed within a test chamber and exposed to low temperature as specified in MIL-STD-810, method 502, procedure II, for a time period of 24 hours. After 24 hours of low temperature exposure, the valve shall be charged to a hydraulic pressure of 3000 psi, and a flow as specified (see 3.4.1) shall be maintained for a period of 4 hours. At the conclusion of the 4 hour period, the valve shall be tested as specified in 4.7.3.7 (50 cycles only). The valve shall then be stabilized at room temperature and tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.7.3.10 Fungus. To determine conformance to 3.4.8.3, the valve shall be subjected to the fungus resistance test specified in MIL-STD-810, method 508. Subsequent to this test, the valve shall be tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.7.3.11 Salt spray. To determine conformance to 3.4.8.4, the valve shall be subjected to salt spray tests as specified in MIL-STD-810, method 509, the exposure period shall be for 100 hours. Subsequent to this test, the valve shall be tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.1.3.11.1 Valve spool corrosion. To determine conformance to 3.4.8.4.1, the valve spool shall be removed from the assembly and subjected to the corrosion test specified in ASTM B117 for not less than 30 hours. After testing, valve spool shall exhibit no corrosion.

4.7.3.12 Sand and dust. To determine conformance to 3.4.8.5, the valve shall be subjected to the sand and dust tests, as specified in MIL-STD-810, method 510, procedure I, at a temperature of not more than plus 125°F. Subsequent to this test the valve shall be tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.7.3.13 Shock. To determine conformance to 3.4.8.6, the valve shall be subjected to the shock tests specified in MIL-STD-810, method 516, procedure I, the value of shock shall be 15 units of gravity (g) exerted through a sawtooth pulse for a period of 11 $\pm$ 1 milliseconds. Subsequent to this test, valve shall be examined for evidence of damage and tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

4.7.3.14 Vibration. To determine conformance to 3.4.8.7, the valve shall be securely mounted to a vibration table and subjected to vibration test specified in MIL-STD-202, method 201A, except vibration shall be for 1 hour only at plus 77° $\pm$ 15°F and 30 minutes at minus 65° $\pm$ 5°F in each direction of the three major axes. Upon test completion, the valve shall be examined for evidence of damage and tested as specified in 4.7.3.1, 4.7.3.6, and 4.7.3.7 (50 cycles only).

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The valves covered by this specification are intended to be used to manually select the fluid flow paths in a 3000 psi hydraulic system in order to control cylinder actuation. They are intended for use with the loader-transporter, guided missile (Hawk-Loader).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of valve required (see 1.2).
c. If first article samples are not required (see 3.1).
d. If construction shall be other than as specified (see 3.3).
e. If finish shall be other than as specified (see 3.6).
f. If responsibility for inspection shall be other than as specified (see 4.1).
g. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
h. If inspection temperature conditions shall be other than as specified (see 4.3).
i. If preproduction inspection is required (see 4.4.1).
j. If initial production inspection is not required (see 4.4.2).
k. If testing conditions shall be other than as specified (see 4.7.3).
l. Applicable level of preservation, packaging, packing and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Subject term (key wood) listing.

Hydraulic valve, three-position, four-way, manual
Manual hydraulic valve, three position, four-way

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6.5 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AT

Review activities:
Army - MI
DLA - CS

User activity:
Army - ME

Preparing activity:
Army - AT

(Project No. 2590-A212)

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